

ROHAN KOUNDINYA

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EDUCATION

KLH University, Hyderabad —

CGPA: 9.3 | 2023-2027

B.Tech in Computer Science & Engineering (Hons.)

Specialization: Data Engineering & AI

Sri Chaitanya (Class XII) — High School

91%

Vignan Vidyalaya (Class X) — High School

93%

INTERNSHIP EXPERIENCE

ISRO – National Remote Sensing Centre (NRSC), Hyderabad May 2025 – Present

AI & Backend Development Intern

- Developed and maintained scalable backend architecture for ISRO's Workflow Recovery Dashboard by designing RESTful APIs, service layers, DAO components, and Oracle database schemas — and integrated a RAG-based AI assistant enabling natural language querying of satellite workflow error logs, reducing manual multi-parameter status lookups to a single conversational query
- Optimized Oracle 21c XE database performance by processing 10,000+ satellite workflow error logs, enabling efficient SQL querying, real-time fault detection, backend monitoring, and reliable data processing across acquisition pipelines.

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL

Backend: Spring Boot, RESTful APIs, Microservices, MVC Architecture, API Integration, Distributed Systems

Databases: PostgreSQL, MongoDB, Oracle 21c, MySQL

AI / ML: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, CNN, BiLSTM, GRU, Vector Embeddings, Semantic Search,

Cloud & DevOps: AWS, Docker, Git, GitHub, n8n, Cloud Deployment, CI/CD, Monitoring, Logging, Kubernetes

PROJECTS

AI-based Sleep Stage Classification using CNN-BiLSTM Attention Network 2026

Python, TensorFlow, Deep Learning

- Achieved 93.77% multi-class accuracy on Sleep-EDF using a proposed attention-based CNN-BiLSTM — outperformed CNN, LSTM, and GRU baselines across 5 sleep stages; evaluated with precision, recall, F1-score, and confusion matrix.
- Engineered a complete EEG preprocessing pipeline: raw signal ingestion → bandpass filtering → 30s epoch segmentation → normalization → class-weighted training with early stopping and adaptive LR scheduling to counter severe class imbalance.

Creator Platform — Scalable Backend System 2024

Java, Spring Boot, MySQL, PostgreSQL

- Architected scalable 15+ Spring Boot RESTful endpoints supporting authentication, profile management, and analytics — RBAC enforcement and structured input validation reduced invalid data entries by 30%, strengthening overall API security posture.
- Optimized MySQL/PostgreSQL query execution and indexing strategies, cutting average response latency by 25% under concurrent load; backed by 100+ unit and integration tests with documented API specifications.

AI Chess Engine — Minimax with Alpha-Beta Pruning 2026

Java, Algorithms, Docker

- Engineered a Minimax engine with Alpha-Beta pruning achieving 8-ply depth in ~2s per move — 40% search space reduction over naive implementation, validated across 100+ gameplay simulations with systematic edge-case testing.
- Containerized the AI Chess Engine using Docker to enable portable, reproducible deployments and consistent execution across environments, improving development, testing, and deployment reliability.

Backend Automation — Contest Data Pipeline 2025

Python, Web Scraping, Google API, Cron

- Engineered an automated scraping pipeline extracting and processing 500+ contest records/cycle from 3+ sources via cron-based scheduling; Google Mail API integration drove a 40% improvement in notification delivery over manual workflows.
- Implemented structured error handling, logging, and regression testing protocols — reducing runtime failures by 35% and ensuring pipeline reliability across iterative feature deployments.

ACHIEVEMENTS & CERTIFICATIONS

- Google College Ambassador** – Conducted technical sessions and coordinated developer outreach initiatives on campus.
- Led a cross-functional team to secure **9th place nationally** (Top 10 All India) in **Mercer VISIONARY Hackathon 2.0** by developing an innovative Augmented Reality (AR)-based solution.
- NVIDIA** – Agentic AI with Large Language Models (LLMs) | Certified Advanced Automation Professional | MongoDB Associate DBA
- AWS Solutions Architect** | **Ai Practioner** | **Cloud Practioner**
- Oracle OCI Gen Ai Professional** | **AI Autonomous DB Professional** | **OCI DevOps Professional (2025)**